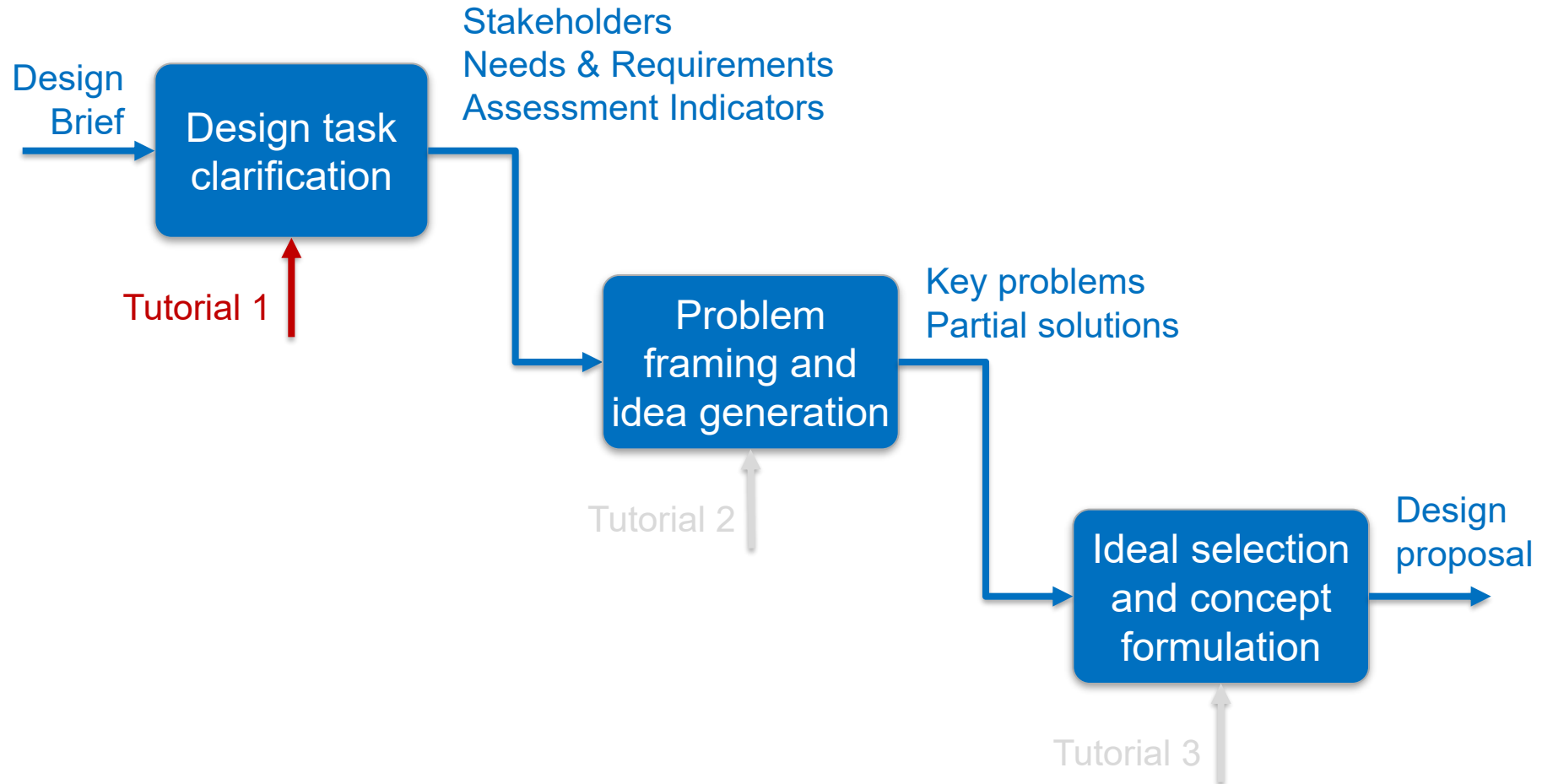


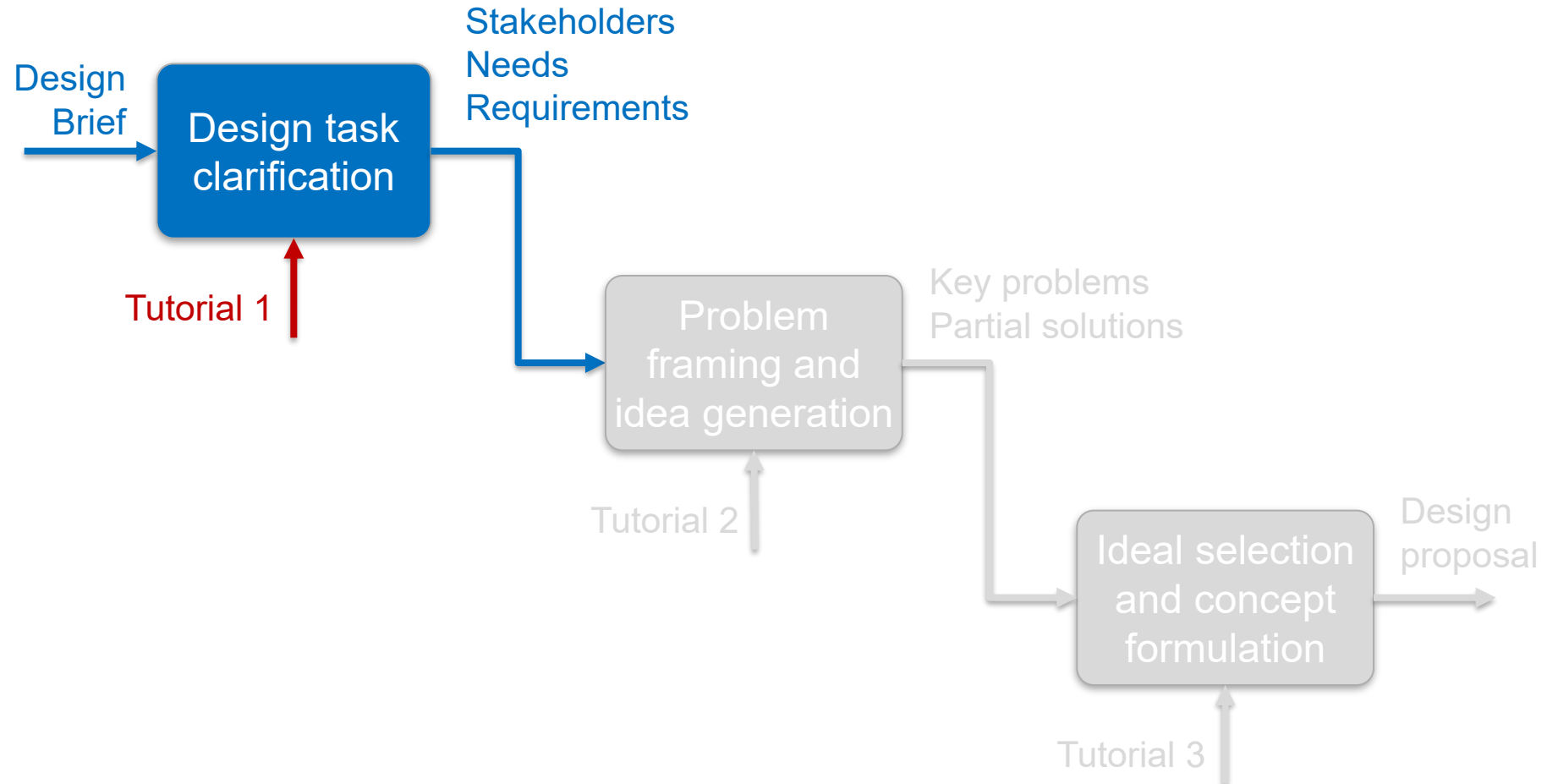
22nd cycle - Spring School 2026
Design Methods and Processes
Tutorial 1
14 March 2026

Gaetano Cascini

DESIGN METHODS AND PROCESSES



DESIGN METHODS AND PROCESSES





DESIGN TASK CLARIFICATION – WHY IT’S SO IMPORTANT?

New requirements that emerge in a late design stage result in skyrocketing costs

- It is extremely advantageous to put all the facets of the design task on the table

The solutions that come out in an instinctive and intuitive way are seldom the most effective

- This applies also to experienced designers and experts in the domain of the design brief
- Creativity is welcomed, but it differs from improvisation substantially

Problems and opportunities are inherently linked

- The clarification of the task and the analysis of the situation help frame the design space of possible solutions, as well as to get rid of non-existing constraints
- A comprehensive definition of the problem “naturally” addresses the designer towards good solutions

STATEMENT (COUNTER BRIEF)

STAKEHOLDERS

NEEDS & REQUIREMENTS



STATEMENT (DESIGN COUNTER BRIEF)

*The design team **agrees** on the formulation of an articulated **statement** for the design task in response to the initial request from the “client”. This step enables the indication of a frame of reference.*

*The **clarification** process allows the design team to be aligned.*

*The purpose of a design counter brief is to demonstrate **how** the designer intends to address the client's needs and goals while offering their own professional insights, expertise, and creative vision. It allows the designer to clarify their **understanding** of the project, present alternative **concepts** or **strategies**, and **justify** their design decisions.*

Answer the questions in the next slides to draft an effective statement.



STATEMENT (DESIGN COUNTER BRIEF)

What is the opportunity, the problem, the challenge?

*Who is your client? What is the “problem” to be addressed?
What is the (time/space) context of the problem?*

What is the target to reach? What is the design goal?

*What are you requested to design to overcome the “problem”?
Are you requested to design a product, a service or a combination
of them? What are your “degrees of freedom”?
To what extent are you constrained?*

STATEMENT (DESIGN COUNTER BRIEF)

A counter brief is a revised version of the client's design brief prepared by the designer after analysing the request

- Clarifies misunderstandings
- Refines vague objectives
- Challenges incorrect assumptions
- Reframes the real problem

It ensures that both client and designer agree on:

- The actual problem to solve
- The goals and constraints
- The target audience
- The success criteria

STATEMENT

Example from ASP Winter & Spring School 16th cycle

DESIGN STATEMENT

THE CLASSROOM OF THE FUTURE

Following the shift in the paradigm of education (methods and spaces) triggered by the pandemic of COVID-19, the goal of the project is to imagine one or more **marketable solutions** that answer the question "what will be the **classroom of the future?**", addressing to some **physical and digital requirements**, such as: customization and modularity, suitability for disabled users and flexible system which allows to build different configurations; moreover, the classroom should be supported by: a responsible technology that allows students to develop an healthy relationship with technology, in-class and out-class communication in order to bring added value to a new learning and teaching experience. All of this have to be possible considering some **constraints**, such as: the budget of a public University, the **sustainability** of materials and processes, and it has to be suitable for existing university structures and different working areas. In terms of the user experience, the solution should foster creativity and physical activity, preventing digital addiction and avoiding distraction, engaging and helping students to collaborate, using technologies to reach an **holistic education** and obtain a citizen who is able to live in the society and effectively contribute in it.

Try to keep the statement within 300 words (even shorter!)

problem **STAKEHOLDERS**

INTERACT WITH THE DESIGN

BENEFIT FROM THE DESIGN

**ARE AFFECTED BY
THE SOLUTION**

PEOPLE

ORGANISATIONS

INSTITUTIONS

GOVERNMENTS

COMPANIES

...

The goal of Stakeholder identification is to understand:

- Who has decision-making power
- Who will use / benefit / interact with the solution
- Who might resist or support it
- Who is indirectly affected

STAKEHOLDERS

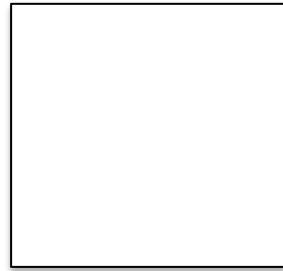
Example from ASP Winter & Spring School 16th cycle



**BE AWARE THAT
DISTANCE FROM THE
DESIGN AND DEGREE
OF INFLUENCE ARE
NOT THE SAME!!**

While it is easier to recognize stakeholders in relation with the design, afterwards it is convenient to rank the stakeholders based on their potential influence

STAKEHOLDERS



ID CARD:

Name:

Characteristics: ...

Role played: ...

Behaviour: ...

Needs: ...

Frustrations: ...

Expectations: ...

Persona cards



Try to **anticipate** the **needs**, **frustrations**, **expectations** of your stakeholders in different **moments** and **circumstances**

Sometimes it is convenient to associate **multiple persona cards** to a single stakeholder

The benefit of representing Stakeholders through Persona cards:

- Humanize stakeholders
- Make them easier to understand
- Keep their needs visible during design decisions
- Encourage empathy

Persona cards help teams stay user-centred and stakeholder-aware throughout the project.

STAKEHOLDERS

The identification of the stakeholders could be an “energy-intensive” task!
On the other hand, recognizing **influential stakeholders** is a key step to ensure success of the design outcome



Focus on stakeholders that
might have different
expectations with respect to
those you already identified



NEEDS

STAKEHOLDERS

*What changes brought on by the designed product/service **are expected** or **should be avoided** in the stakeholders' perspective?*

Consider also the
lecture by Deborah Nas

PEOPLE
ORGANISATIONS
INSTITUTIONS
GOVERNMENTS
COMPANIES
.....

STAKEHOLDERS' PROFILES

Example from ASP Winter & Spring School 16th cycle



Name: Sara
Age: 24 age old
Nationality: Italian
Occupation (role): MSc student of Interior design



Name: Luke
Age: 24
Nationality: English
Occupation (role): Bachelor student in Communication design



Name: Carlo
Age: 60
Nationality: Italian
Occupation (role): Architecture professor



Name: Gustavo
Age: 63
Nationality: Spanish
Occupation (role): Director



Name: Sandro
Age: 45
Nationality: Italian
Occupation (role): Cleaner

About

He is working in a cleaning company and is involved in cleaning the university and all its stuff and spaces.

Needs

- He needs a time during the day to clean the classes and the bathrooms so his job becomes really time consuming
- He has to work more due to COVID-19 pandemic

Frustrations

- Students are always around while he is doing his job so he has to do it in not daily time (after 7PM).
- He is worried to lose his job if the education system completely shifts online. But at the same time he does not want his workload to be increased too much because of the pandemic.

Expectations

- He would like his work to be respected both from students and university staff.
- He would like university students to be more careful about the environment he is daily cleaning.
- He would like a less time consuming job and more organized thanks to the less frequented spaces.

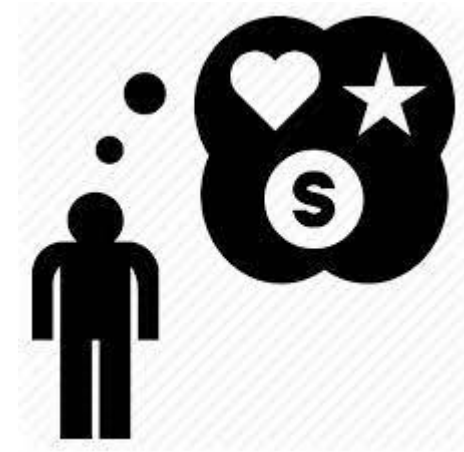
Frustrations

- Technology plays a
- it is difficult to com down

- Possibility to hold both theoretical and practical classes online

STAKEHOLDERS' NEEDS

The identification of latent needs opens up to the creation of new value and innovation opportunities



Don't limit your exploration to
the product/service in use.
Consider its entire life cycle!

A guide to capture NEEDS & REQ.TS

Human-Based

End-user

-HE-

They set how the product/service should “positively” influence **humans’ quality of life** i.e. their activities and mental status.

Technical

Functional

-TF-

They set the “must-have” **performance** and the **overall quality/efficiency** of the product/service you will design.

Business

Corporate

-BC-

They set the **cost/financial/market-based** characteristics of the solution whose fulfilment is fundamental to make profit and to be compliant with the investors’ business mission.

Regulatory

Policy

-RP-

They set the aspects of the product/service that have to be validated with respect to **rules** and **standards**.

STAKEHOLDERS' NEEDS

Example from ASP Winter & Spring School 16th cycle

Tech/funct

- Safety
- Time flexibility
- user friendliness
- laboratory viability
- collaboration and social contact
- physical activities
- Self explanatory tools
- remote working flexibility
- Inclusive HMI
- social interactions
- physical accessibility
- Shared space availability
- Fast implementation and deployment

Human based

- Safety
- Physical and psychological support
- Privacy
- user friendliness
- Collaboration and social contact
- Real time debate
- new digital skills
- environmental consciousness
- physical activities
- remote working flexibility
- Inclusive HMI
- physical accessibility

Business

- Economical support for students
- Fast implementation and deployment
- Self-sustainability of the investments

Regulatory policy

- Privacy
- laboratory viability
- environmental consciousness
- physical accessibility

from NEEDS & REQ.TS to measurable INDICATORS

*How will the fulfilment of needs
and requirements be evaluated?*



Tip!

- Needs are usually expressed in a **narrative way**: we have thus to translate them into **something measurable** (i.e. an **INDICATOR**), so that we can clearly demonstrate whether the solution we have designed is able to fulfil them

FROM NEEDS TO REQUIREMENTS

Example from ASP Winter & Spring School 16th cycle

NEED	STAKEHOLDERS	INDICATORS
Allow differentiated and customizable Kinds of Lecture and Laboratories (include frontal lecture - lab lecture - reviews)	• practical professor • external company • rector (high quality)	• number of different supported platform and lecture formats
Lean Exam Management	• practical professor • theoretical teacher	• number of reported cheating events

Direct contact with companies	• student • rector • foreign student • external company	• % of students attending company fairs • Number of project works offered by companies in regular courses
Handly management of physical devices	• theoretical professor • student	• Set-up time to start a lecture • SUS (System Usability Scale)

NEEDS AND REQUIREMENTS



Tip!

Design specifications can be extremely long. You are advised to **cluster needs and requirements** into thematic areas to handle them more efficiently.



Tip!

Different stakeholders may share the same need. Different stakeholders may have **seemingly conflicting** needs. These needs and the **corresponding indicators and expected values** are to be defined very carefully.



Tip!

By definition, a **stakeholder** has at least a need linked with the design. Check how you have filled in the “expectations” field of the personas.

NEEDS AND REQUIREMENTS



Tip!

It may happen that **new stakeholders** “**appear**” during the requirements formulation step. Iterations are common in design, don’t panic, but don’t abuse, because this makes the design process poorly efficient..



Tip!

Organize the needs into a **hierarchy** of primary, secondary and tertiary needs to set **priorities** in your design process



Tip!

The **same category of stakeholders** includes elements with many **peculiarities**. Try to identify the **distinguishing traits** that are the most relevant for your design task.



Tip!

It may happen that some stakeholders have **different needs** in **different moments** and circumstances.



NOW IT'S YOUR TURN: clarify your design task!

1. Listen to the **CHALLENGE** presented in the next few slides. Later you will work in teams → Read the **DESIGN (COUNTER)BRIEF** carefully
2. Identify and describe your **STAKEHOLDERS** (e.g. with a set of “Persona” cards, highlighting degree of influence and potential conflicts)
3. Define the list of specifications (including hierarchy of **NEEDS/REQUIREMENTS** to address, related priority and **ASSESSMENT INDICATORS**)
4. Be ready to **PRESENT** what you have done in less than **5 MINUTES** to the entire classroom (**NO SLIDES**)

NOTE: This will be the first activity of your ASP project to be completed before the residential week in OROPA (more info on April 18)

22nd cycle - Spring School 2026
Design Methods and Processes
Today's challenge
14 March 2026

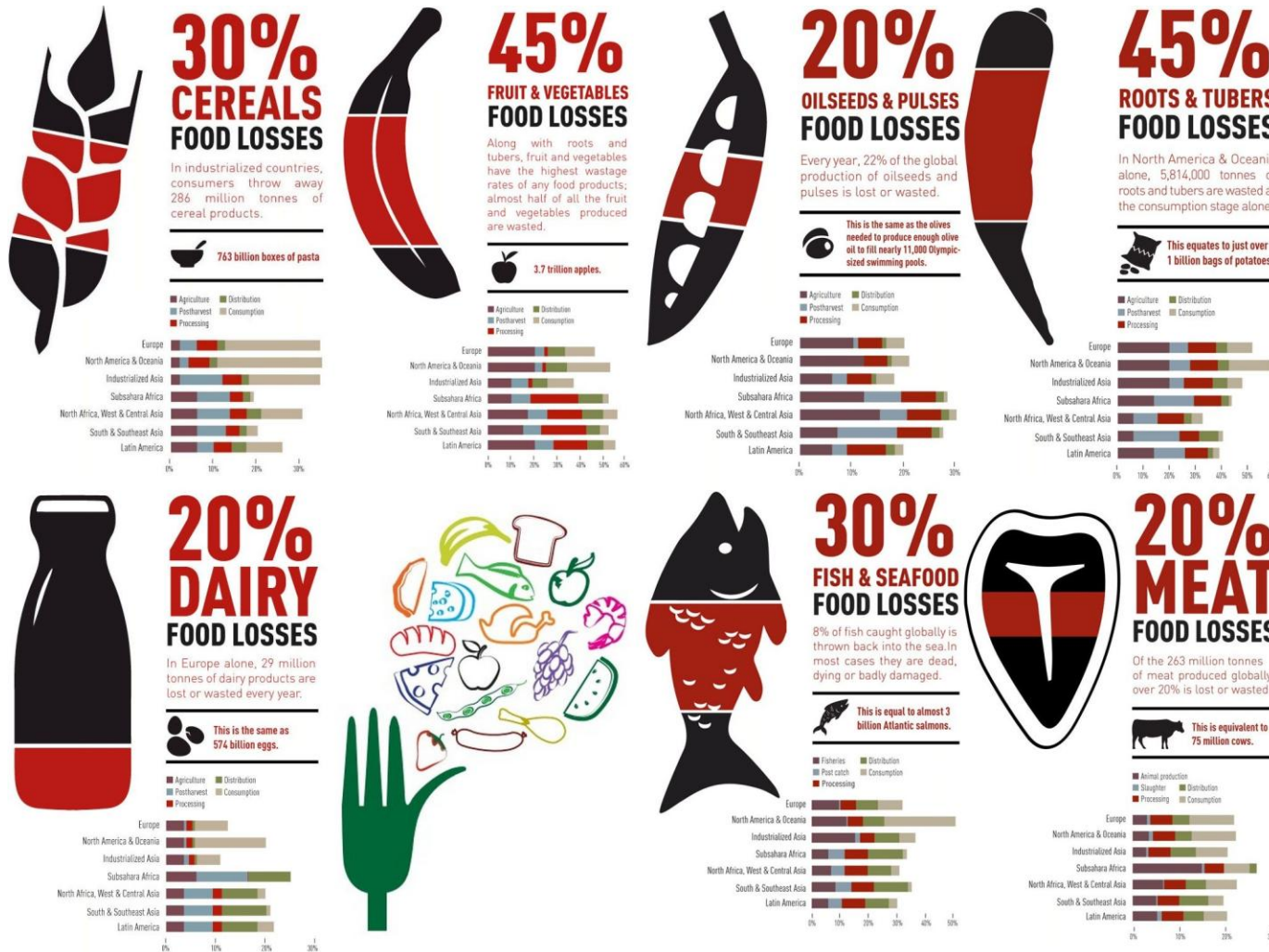
Gaetano Cascini

CONTEXT – FARM2FORK STRATEGY

The Farm to Fork Strategy is at the heart of the **European Green Deal** aiming to make **food systems fair, healthy and environmentally-friendly**



CONTEXT – FARM2FORK STRATEGY

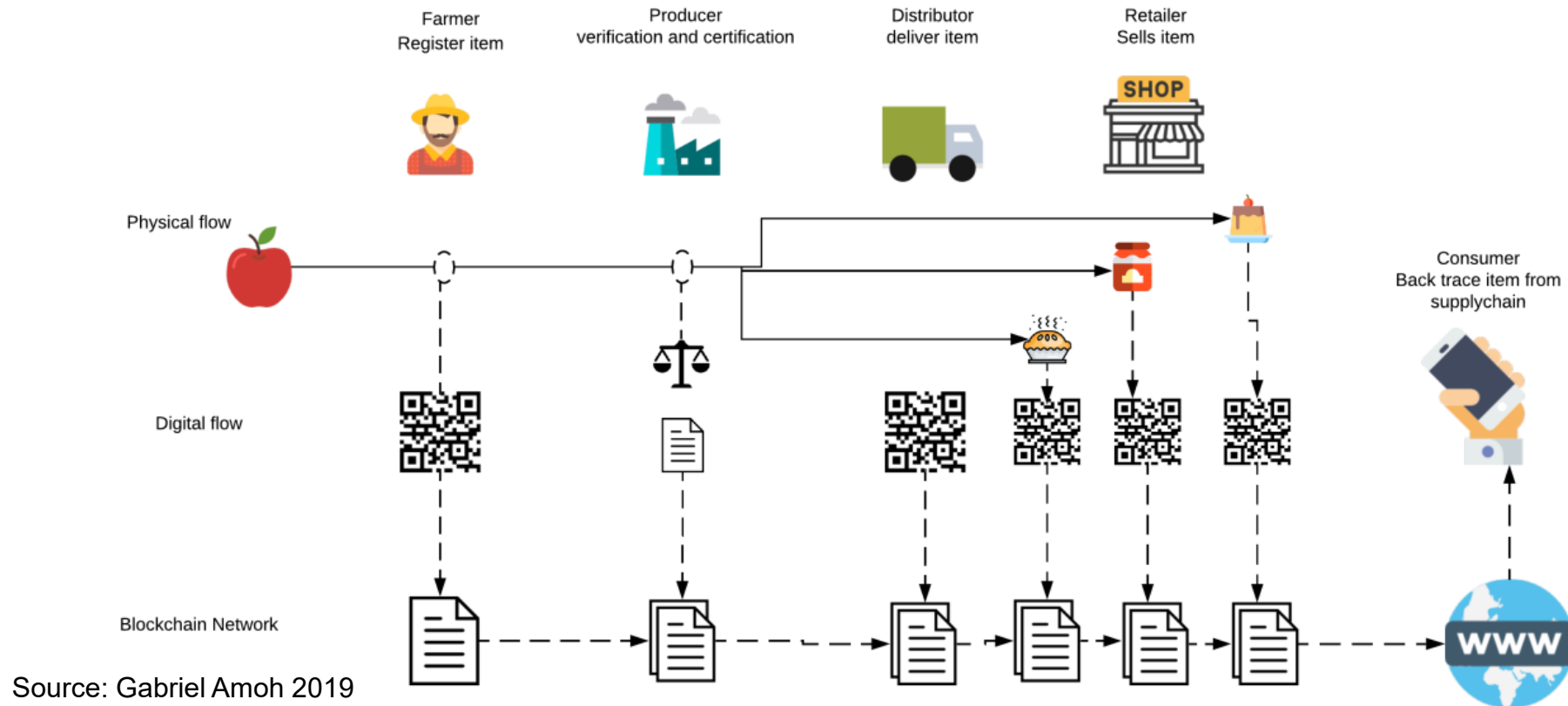


There is a need for more **data-driven decision making**, even though consumers are sceptical to share data, least to the government.

Better understanding of **citizens' food consumption behaviour**, the factors influencing **choices** and drivers that would facilitate **changes in behaviour** in an **inclusive manner** towards healthy and sustainable food consumption practices.

World Food Waste. Image credit: Infographics by FAO

CONTEXT – FARM2FORK STRATEGY

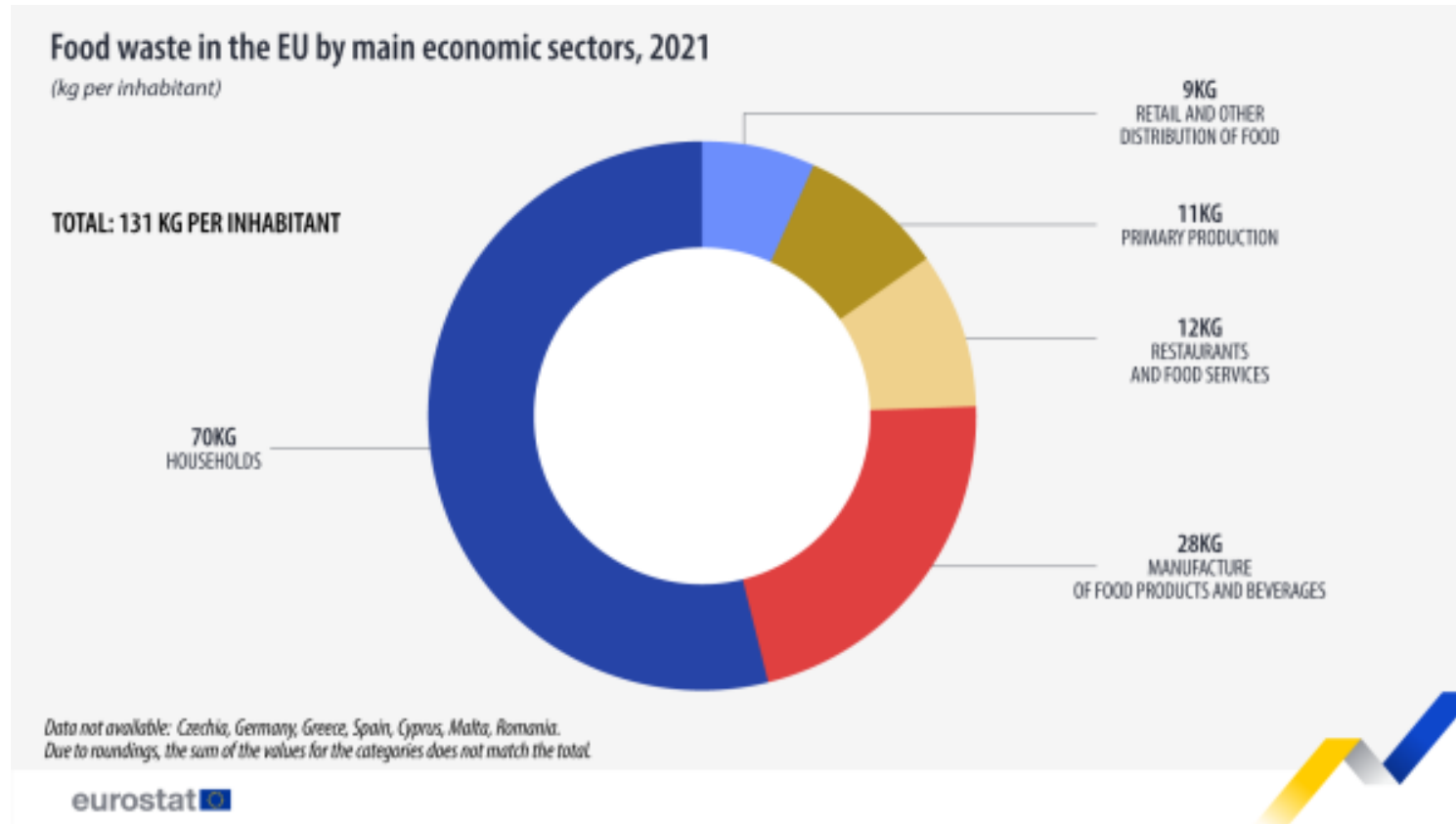


Many solutions have been developed during the years aiming at ensuring food traceability. These solutions leverage AI and blockchain in order to **monitor the path followed by food from farm to supermarkets' shelves** or to optimise food logistic operations.

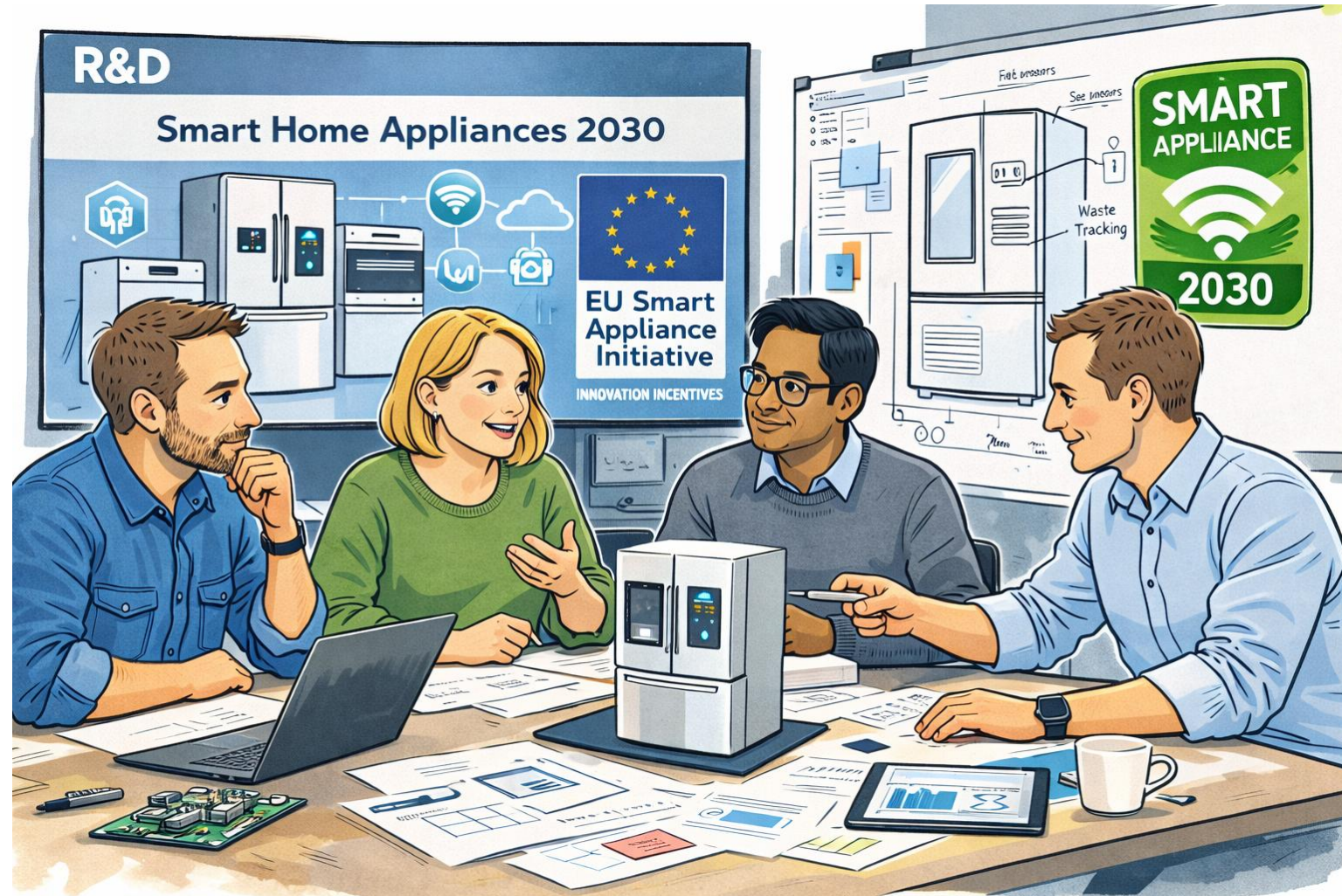
CONTEXT – FARM2FORK STRATEGY

Large portion of **food is wasted at home**.

However, until now few initiatives have been launched in order to **close the gap between supermarket's shelves and what happens at home**.



CONTEXT – HYPOTHETICAL SCENARIO



Your team is part of the **R&D Dept** of a household appliance manufacturer.

A **new European initiative** is going to incentivize the adoption of household appliances labelled “**smart**”, i.e., capable of collecting data related to food conservation, preparation, consumption, and waste.

Your company intends to release by 2030 a new series of appliances named “**Kitchen-Sense**” suitable to be labelled as “smart” according to the novel labelling system.

THE DESIGN CHALLENGE



How to collect data on food conservation, processing, consumption and waste in private homes in an inclusive manner without violating the privacy of citizens?

THE DESIGN CHALLENGE

Explore **which data types are most useful** to share (behavioural data, data from private providers, such as data gathered by relevant apps, stated data...etc.) and how to meaningfully harmonize data to use data for food system transformation by different actors, and which tools to best make use of, such as Artificial Intelligence (AI) while analysing how consumer data can be shared in an anonymized and safe way complying with the **General Data Protection Regulation (GDPR)** rules.



What are the relevant stakeholders in this challenge?

What are their needs and expectations?

How to assess their fulfilment?



THE DESIGN CHALLENGE BRIEF

The European Green Deal and its Farm to Fork Strategy call for a transition toward food systems that are fair, healthy, and environmentally sustainable.

While significant progress has been made in improving food traceability along the supply chain—through AI, blockchain, and advanced logistics—much less attention has been paid to what happens after food reaches the supermarket shelf. A substantial proportion of food waste occurs in private homes, where consumption patterns, storage behaviours, and everyday decisions strongly influence environmental and social outcomes.

European institutions are moving toward incentivizing the adoption of smart household appliances capable of collecting data related to food conservation, preparation, consumption, and waste. By 2030, large incentives may be introduced for consumers who purchase such appliances. However, the collection and use of domestic data raise significant challenges related to privacy, trust, inclusivity, interoperability, and compliance with the General Data Protection Regulation (GDPR). Consumers are often sceptical about sharing personal data—particularly with public authorities—yet more granular and reliable data are essential to enable evidence-based policymaking and systemic change.

The task is to frame the design specifications of this product-service system. How can appliances collect meaningful data on food storage, processing, consumption, and waste inside private homes? Which types of data would be most valuable for different actors in the food system? How can data be harmonized and analysed while ensuring anonymization, security, inclusivity, and full regulatory compliance?

CONCLUDING REMARKS

The challenge was conceived as a design task with a similar degree of complexity to the ASP multidisciplinary projects

- We don't expect that you generate your solutions in less than two hours!
- We actually don't expect you to generate any solution at this stage!

Focus on the exploration of the design space beyond the brief you have been assigned

- Get familiar with the techniques we just introduced in the Tutorial 1 session
- Recognize stakeholders, their needs and expectations, and the indicators you could use to assess the quality of your design

Expected outcome

- Learn how to structure your design task exploration to guide your process in the ASP multidisciplinary project you will be assigned in April (once we meet in the residential week in Oropa, this activity must be completed already)

Teams A-J → 4V

Teams K-T → 6V

Shared Folder:

